

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE BIOSYSTEMS AND CIVIL ENGINEERING

ECU 108: INTRODUCTION TO ENGINEERING PROFESSION

DATE: Friday 9th June 2017

TIME: 8.00a.m-10.00a.m

INSTRUCTIONS:

This paper consists of two parts; Part A for BSC Civil Engineering and Biosystems and Agriculture Engineering. Part B for BSC Electrical and Electronic Engineering; BSC Biomedical Engineering; and BSC Petroleum Engineering

PART A

Students are required to attempt question one and any other two questions

Question 1

a) Outline the full naming of the following abbreviations and state their significance;

- i) EBK
- ii) CUE
- iii) SHE
- iv) PPE
- v) PPP



(15 marks)

b) List the FIVE subject areas that an Engineering student needs to cover. For each case, give TWO course units studied.

(10 marks)

c) State the main sources of pollutants in Nairobi river

(5 marks)

Question 2

a) With the aid of flow diagram explain the main in need analysis

(10 marks)

b) Discuss FIVE steps necessary in solving a design problem.

(10 marks)

Question 3

a) State the 4R principles.

(4 marks)

b) State FOUR types of renewable energies

(4 marks)

c) Waste management and the Green revolution are not well adopted in Kenya. Explain the drawbacks and opportunities towards these scenarios

(12 marks)

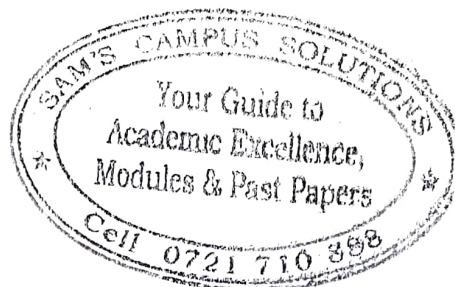
Question 4

a) Briefly describe the evolution of the Engineering discipline and profession. (10 marks).

b) Describe the process of registering as a Professional (Practicing) Engineer. (10 marks)

PART B

Answer questions ONE and any TWO other.



1. (a) What is engineering profession?

(3marks)

i. Explain how the role of an electrical OR a mechanical engineer in government (ministry) differs from that of a counterpart in a chemical manufacturing facility.

(5marks)

ii. Describe the responsibility of an engineer to the public.

(4marks)

(b) Water, roads and power supply are essential for economic development of a country.

i. Explain the specific role of electric power in the economic development of Kenya.

(4marks)

ii. Describe the engineer's contribution in this process.

(4marks)

(c) Describe the role of an engineer in a disaster which may result in flooding after unusually heavy rains.

(5marks)

(d) Explain why it is important for the engineer to do accurate data collection at the start of any project.

(5marks)

2. (a) Describe the work done by an engineer in one discipline, including the context in which that work is done.

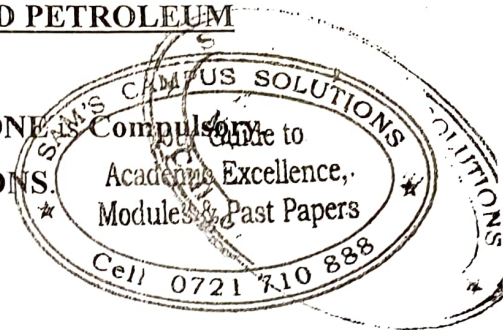
(6marks)

(b) Describe the role of an engineer in making water useful in modern society. (8marks)

SECTION B

TO BE ATTEMPTED BY ELECTRICAL, BIOMEDICAL AND PETROLEUM
ENGINEERING STUDENTS ONLY

- 1) This Section contains a total of **FIVE** questions. Question **ONE** is **Compulsory**.
- 2) Answer **QUESTION ONE** and any other **TWO QUESTIONS**.



QUESTION 1 (30 marks)

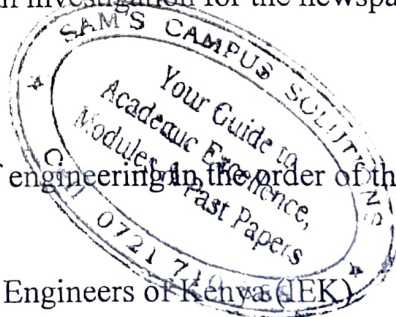
- A. Alfie, a renowned structural engineer, is hired for a nominal sum by a large city newspaper to visit the site of a state bridge construction project, which has had a troubled history of construction delays, cost increases, and litigation primarily as a result of several well-publicized, on-site accidents. Recently the state highway department has announced the date for the opening of the bridge. State engineers have been proceeding with repairs based upon a specific schedule. Alfie visits the bridge and performs a one-day visual observation. His report identifies, in very general terms, potential problems and proposes additional testing and other possible engineering solutions.

Thereafter, in a series of feature articles based upon information gleaned from Alfie's report, the newspaper alleges that the bridge has major safety problems that jeopardize its successful completion date. Allegations of misconduct and incompetence are made against the project engineers and the contractors as well as the state highway department. During an investigation by the state, Alfie states that his report was intended merely to identify what he viewed were potential problems with the safety of the bridge and was not intended to be conclusive as to the safety of the bridge.

Question:

Was it ethical for Alfie to agree to perform an investigation for the newspaper in the manner stated? (15 marks)

- A. State & explain *four* phases of the history of engineering in the order of their occurrence. (10 marks)
- B. List any four objectives of the Institution of Engineers of Kenya (IEK). (5 marks)



QUESTION 2 (20 marks)

- A. State and explain any *five* key functions of engineering societies. (10 marks)
- B. Outline any *four* provisions addressed in the "Engineers Act 2011" of Kenya. (6marks)
- C. State four factors that influence Professional Ethical standards in engineering. (4 marks)

QUESTION 3 (20 marks)

- A. State and explain any *five* key functions of engineering societies citing example in each. (10 marks)
- B. Engineers in Kenya are *key* to quality infrastructure. Explain. (6 marks)
- C. What do you understand by the term “*ethical dilemma*” and how can it be solved? (4 marks)

QUESTION 4 (20 marks)

- A. With relevant examples, explain how engineers impact the lives of Kenyans on a day to day basis? (10 marks)
- B. List *three* differences between law and ethics. (6 marks)
- C. State *four* major requirements of a professional engineer in Kenya? (4 marks)

QUESTION 5 (20 marks)

- A. ‘Engineers are social experimenters’, justify. (15 marks)
- B. Briefly explain how the Institution of Engineers of Kenya was formed. (5 marks)

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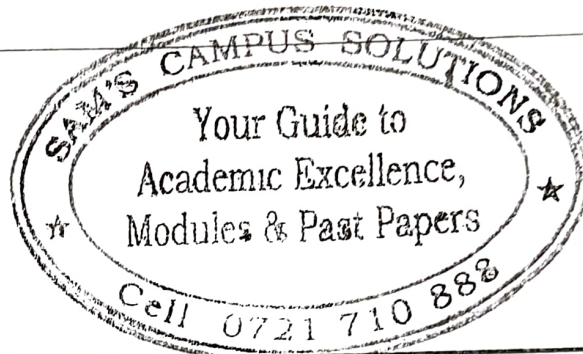
CONTINUOUS ASSESSMENT TEST (CAT) I

ECU 108: INTRODUCTION TO THE ENGINEERING PROFESSION

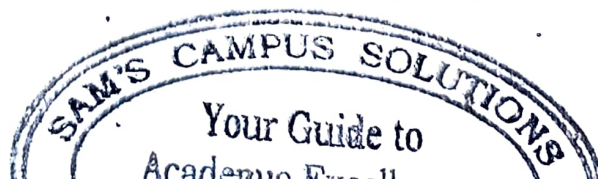
1st March 2019

INSTRUCTIONS:

- Answer All Questions.
- Define All Symbols Used.
- All Questions Carry Equal Marks.



- a)
 - (i) State what *Engineering* is.
 - (ii) Give a brief history of Engineering during the Ancient era, Renaissance era, and Modern era.
- b)
 - (i) Discuss *Engineering and Science*.
 - (ii) Write an essay on *Civil Engineering* giving its definition, sub-disciplines and scope; as well as the major specialities.
- a)
 - (i) Give a general definition of a *Professional Engineer*.
 - (ii) Outline the steps towards becoming a Professional Engineer.
- b)
 - (i) Give the main branches of Engineering and their areas of focus.
 - (ii) Outline the challenges for engineers in society.
- a)
 - (i) Explain what the Building code is.
 - (ii) "Building Works" is a technical term under the Building Code. Explain its meaning.
- b)
 - (i) Outline the stages of *Building and Civil Engineering Works*.
 - (ii) Give the primary responsibility of the different consultants involved in Building Works.



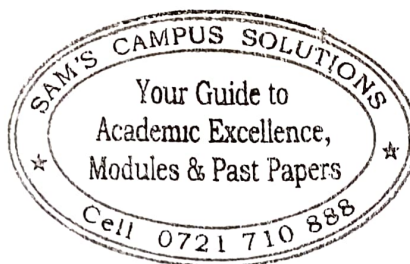
**TO BE ATTEMPTED BY CIVIL ENGINEERING AND BIOSYSTEMS STUDENTS
ONLY**

1) This Section contains a total of **FIVE** questions. Question **ONE** is **Compulsory**.

2) Answer **QUESTION ONE** and any other **TWO QUESTIONS** in this Section.

QUESTION ONE

- a) What is engineering? And who is an engineer? (6 marks)
- b) Explain what a profession is. (2 marks)
- c) List and state clearly all major code of ethic fundamentals known as Canonis. (15 marks)
- d) Write short notes on:
- 1) Philosophy (2 marks)
 - 2) Religion (2 marks)
 - 3) Virtues/ethics (2 marks)
 - 4) Existence of right and wrong (2 marks)
 - 5) Habits and morals (2 marks)
 - 6) Developing a model of a person showing components of the psyche (4 marks)
- e) List and state clearly what engineers need to know. (4 marks)
- f) What are engineering tasks? (4 marks)



QUESTION TWO

- I) Discuss what emerging engineers need to know in practice. (5 marks)
- II) Discuss what you are supposed to do within 5 years as an engineer. (10 marks)
- III) Explain what is the role of code of ethics. (5 marks)

QUESTION THREE

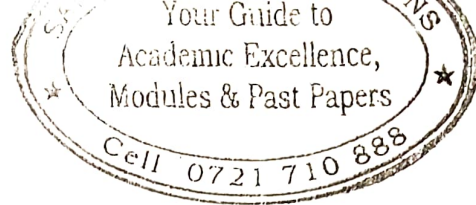
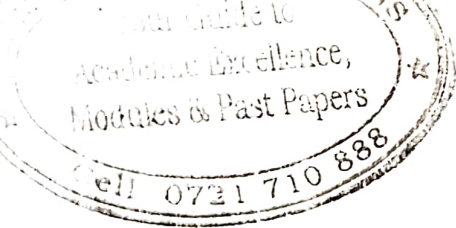
Engineering is becoming increasingly specialized. Discuss. (20 marks)

QUESTION FOUR

- I) Give an account on code of ethics for engineers (8 marks)
- II) Engineers mostly work in the business world. Discuss. (12 marks)

QUESTION FIVE

Demonstrate procedure for solving ethical conflicts for both internal appeal option and external appeal option. (20 Marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE

ECU 108: INTRODUCTION TO THE ENGINEERING PROFESSION

DATE: FRIDAY 12TH APRIL 2019

TIME: 8.00A.M. - 10.00A.M.

INSTRUCTIONS:

- (i) There are **Five** questions.
- (ii) Question No.1 is **Compulsory** and carries **30 Marks**.
- (iii) Select any **other Two** questions, each of which carries 20 Marks.
- (iv) Define all symbols used.
- (v) Marks will be awarded for procedure, methodologies used and not necessarily on the final answer.

- Q1. (a) (i) Describe how the construction of buildings is regulated in Kenya. (5 Marks)
- (ii) Give five operations which are deemed under the Building Code as erection of a building. (5 Marks)
- (b) (i) Distinguish between a feasibility study and a strategic plan. (4 Marks)
- (ii) State the role a manager plays in an industrial organisation and briefly explain how they do so. (6 Marks)
- (c) (i) Explain what chemical engineering is, discuss its advent and what led to its development. (5 Marks)
- (ii) Outline the sub-disciplines of mechanical engineering and give their scope. (5 Marks)
- Q2. (a) Explain the terms contract drawings and built drawings. (4 Marks)
- (b) Describe what engineering activity the following would be engaged in:
- (i) An electrical engineer deployed in power. (8 Marks)
- (ii) A mechanical engineer engaged in maintenance and operations. (8 Marks)

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- Q3. (a) Under which law is the practice of engineering regulated in Kenya? (4 Marks)
(b) Name the four main engineering branches and state their focus. (16 Marks)
- Q4. (a) State the critical factors that influence the location of industrial projects. (4 Marks)
(b) Name the vital management functions for sound management of organisations and briefly explain what each entails. (16 Marks)
- Q5. (a) Explain what a fully fledged feasibility study is and its role in the implementation of industrial investment projects. (6 Marks)
(b) What is depreciation and why is it necessary in industrial projects? (6 Marks)
(c) Project costs are grouped into fixed costs and variable costs. With the help of relevant examples, explain these terms. (8 Marks)

PART A:

**TO BE DONE BY CIVIL AND BIOSYSTEMS ENGINEERING
CANDIDATES ONLY**

INSTRUCTIONS:

Answer question 1 and TWO others

1. (a)
 - i. Who is an Engineer?
 - ii. How does the Engineering profession differ from other professions?

(3marks)
 - (b) Write short notes to show you understand the difference between each of the following.
 - i. Science
 - ii. Engineering
 - iii. Technology
 - iv. Research
 - v. Innovation

(10marks)
 - (c) Briefly describe the SIX fundamentals Canons of the Code of Ethics for Engineers.

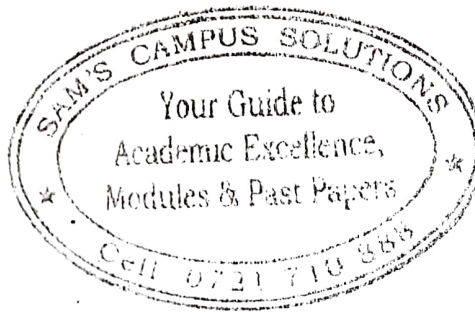
(12marks)
 - (d) Discuss the main engineering components considered in a building development project.

(5marks)
-
2. (a) Under the Engineers Board of Kenya, explain the conditions to be met for registration of :
 - i. A Professional Engineer
 - ii. A Consulting Engineer

(4marks)
 - (b) Briefly outline some SIX contractual forms of employment under which an Engineer may be employed.

(6marks)
 - (c) Describe the general responsibilities of an engineer.

(10marks)



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3. (a) Engineering involves presentation of reports from time to time.
- i) Why is it necessary to prepare such reports? (4 marks)
 - ii) What is the difference between formal writing and informal non-verbal communication? (4 marks)
- (b) Give an example of report writing during the university course undertaken by engineering students. (2 marks)
- (c) When is it necessary to include alternative solutions in an engineering report? (3 marks)
- (d) Give a suitable structure for a formal report given for a problem which has been solved by an engineer or team of engineers. (7 marks)
4. Large projects involve multi-disciplinary teams of professionals, which may include engineers and other professionals.
- a) Give an example of such a project team, citing the name of project you have chosen. (4 marks)
 - b) Who will be the team leader in such a project, and why? (3 marks)
 - c) Apart from the team leader, who else will be responsible for giving reports on the project? (5 marks)
 - d) At the end of the project, describe the reports which will be needed by various actors. (4 marks)
 - e) What happens at the end of such a project. (4 marks)

PART A:

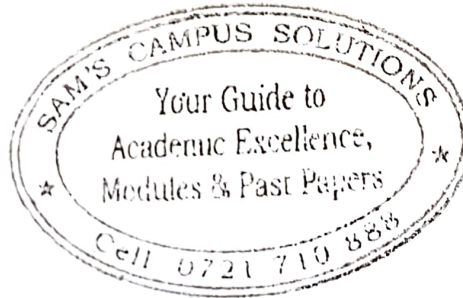
TO BE DONE BY CIVIL AND BIOSYSTEMS ENGINEERING

CANDIDATES ONLY

INSTRUCTIONS:

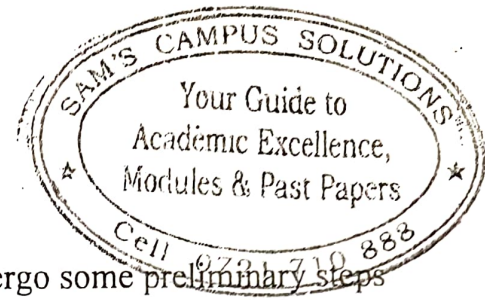
Answer question 1 and TWO others

1. (a)
 - i. Who is an Engineer?
 - ii. How does the Engineering profession differ from other professions?
(3marks)
 - (b) Write short notes to show you understand the difference between each of the following.
 - i. Science
 - ii. Engineering
 - iii. Technology
 - iv. Research
 - v. Innovation
(10marks)
 - (c) Briefly describe the SIX fundamentals Canons of the Code of Ethics for Engineers.
(12marks)
 - (d) Discuss the main engineering components considered in a building development project.
(5marks)
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2. (a) Under the Engineers Board of Kenya, explain the conditions to be met for registration of :
 - i. A Professional Engineer
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(4marks)
 - (b) Briefly outline some SIX contractual forms of employment under which an Engineer may be employed.
(6marks)
 - (c) Describe the general responsibilities of an engineer.
(10marks)



PART B:

**TO BE DONE BY ELECTRICAL, BIOMEDICAL AND PETROLEUM
ENGINEERING CANDIDATES ONLY**



Answer Question ONE and any 2 other questions.

1. (a) Define the term "Engineer". (3 marks)
 - (b) For one to become an engineer, the person has to undergo some preliminary steps before acceptance.
 - i) Describe the steps which are required to be taken. (6 marks)
 - ii) Explain why each of these steps is necessary. (6 marks)
 - iii) Under what circumstances may an engineer be disqualified from this description? (5 marks)
 - (c) Explain the difference between the institution of Engineers of Kenya and the Engineers Board of Kenya, including the role of each body. (6 marks)
 - (d) Explain in your own words why you chose to become an engineer, and which type of engineer. (4 marks)
2. (a) The engineering profession is organized into a number of disciplines, which specialize in particular aspects of the profession.
 - i) Describe the work done by TWO disciplines of engineering. (8 marks)
 - ii) Explain the important problems that the disciplines you have chosen solve. (5 marks)
- (b) In the course of studying for the engineering degree, a number of "common units" are part of the curriculum. Explain why it is necessary to study such units, citing two examples. (4 marks)
examples. (4 marks)
- (c) Why is it important for the engineer to be a person of good character in his private and professional work? (3 marks)

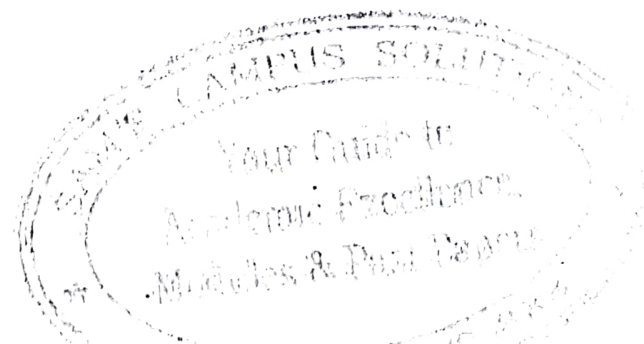
ECU 108 27th August 2018

3. (a) Using roads as an example, discuss the engineering tasks, including data collection, analysis, planning design, implementation, operation, maintenance of an engineering project in a rural county. (8marks)

(b) Describe at some six Professional Obligations of an Engineer (12marks)

4. (a) Describe Four Strategies for promoting science, technology & innovation under the Kenya Vision 2030. (8marks)

(b) Using dam construction as an example, discuss the considerations for the utility, safety, health, ergonomic and the environment in the planning and implementation of an engineering project. (12marks)



KENYATTA UNIVERSITY
FIRST YEAR, SECOND SEMESTER 2017/2018
BACHELOR SCIENCE IN ^{AGRICULTURAL} ENVIRONMENT AND BIOSYSTEMS ENGINEERING
AND
BACHELOR SCIENCE IN CIVIL ENGINEERING

ECU 108: INTRODUCTION TO THE ENGINEERING PROFESSION

CONTINUOUS ASSESSMENT TEST 2

DATE: 6th August 2018

TIME: 1 Hour

INSTRUCTIONS

Answer All the Questions

1. (a) Write short notes to show you understand the difference between each of the following
 - (i) Science and Research
 - (ii) Engineering and Technology
 - (iii) Creativity and Innovation

(6 marks)
- (b) For engineering projects, distinguish between:
 - (i) Long term vs. short term Agreements
 - (ii) Exclusive and Non-exclusive Agreements

(4 marks)
2. a) Distinguish between the following:
 - (i) A Graduate Engineer
 - (ii) A Professional Engineer

(4 marks)
- (b) Discuss some six steps for Technology Transfer

(6 marks)
3. a) What is a project Plan?

(2 marks)
- (b) Discuss briefly the four stages of a project management cycle for an engineering project

(8 marks)
- 4) In a building project, what are the considerations for the utility, safety, health, ^{economic} ergonomic and the environment in the planning and implementation of an engineering project?

(10 marks)
5. Describe the role of an engineer in rural development at County level in Kenya

(10 marks)

become a problem to the humans, and what can be done to deal with the problem? (6marks)

The sun is a potential abundant source of energy in Kenya.

(i) How can this energy be used? (4marks)

(ii) In which areas of Kenya would it be useful? (5marks)

b) What are limitations of this as a source of electrical energy? (5marks)

c) What other form of energy can be combined with this to achieve reliable power supply? (6marks)

4. In a building project, the professional team includes architects, quantity surveyors and engineers.

(a) Explain the role of a mechanical engineer OR electrical engineer in the design of a high-rise office building. (5marks)

(b) What is the role of the same engineer:

(i) During the closing stages of the project? (5marks)

(ii) After the project hand-over? (5marks)

(c) What is the difference between the biomedical engineer and the other two disciplines? (5marks)

5. (a) What is the difference between a Chairman and a Chief Executive Officer of a company? (6marks)

(b) Describe the role of the Board of Directors in the management of a government-owned entity. (6marks)

(c) Explain the difference between the Statement of Comprehensive Income (Profit and Loss Statement) and the Statement of Financial Position (Balance Sheet). (8marks)

BACHELOR SCIENCE IN ENVIRONMENT AND BIOSYSTEMS ENGINEERING
AND
BACHELOR SCIENCE IN CIVIL ENGINEERING

ECU 108: INTRODUCTION TO THE ENGINEERING PROFESSION

CONTINUOUS ASSESSMENT TEST 1

DATE: 2nd July 2018

TIME: 2 Hours

INSTRUCTIONS

Answer All the Questions

1. (a) Distinguish between:
 - (i) An aeronautical Engineer and an Aerospace Engineer
 - (ii) A Civil Engineer and a Biosystems Engineer
 - (iii) A Mechanical Engineer and an Industrial Engineer(6 marks)
(b) Discuss the advantages of working in the Engineering Profession (4 marks)
2. (a) How does the Engineering profession differ from other professions? (3-marks)
(b) Describe the major roles and responsibilities of an engineer. (7 marks)
3. (a) Outline some five fundamental Canons of the code of ethics for engineers (5 marks)
(b) Discuss how an engineer facilitates public safety and health (5 marks)
4. (a) Differentiate between:
 - (i) Part-time employment and casual employment
 - (ii) Apprenticeship and internship
 - (iii) Part-time employment and probationary work(6 marks)
(b) For a firm specializing in Engineering infrastructure development, discuss the role of policy in the organization (5 marks)
5. (a) What is organizational culture? (2 marks)
(b) Describe **Four** Strategies for promoting science, technology & innovation under the Kenya Vision 2030. (8 marks)